Bilinear and Quadratic Forms

From the class book: Chapter 8, Sections 8.1–8.5

1. Key definitions

Bilinear form. A map $f : V \times V \to F$, linear in each argument. In a basis $\beta = \{v_i\}$ its matrix is $A = [f(v_i, v_j)]$ and $f(x, y) = [x]_\beta^t A [y]_\beta$. It is *symmetric* if $f(u, v) = f(v, u)$ and *alternating* if $f(v, v) = 0$ for all v.

Quadratic form. $q(x) = x^t A x = \sum a_{ii} x_i^2 + 2 \sum_{i<j} a_{ij} x_i x_j$ with $A = A^t$: the coefficient of $x_i x_j$ is split as half in (i, j) and half in (j, i). Polarization recovers $f(x, y) = \frac{1}{2}(q(x + y) - q(x) - q(y))$.

- **Congruence:** a change of basis P turns A into $P^t A P$. Congruent matrices have the same rank, so the rank of f is well defined; f is non-degenerate iff A is invertible.
- **Inertia:** p and q are the numbers of positive and negative eigenvalues. Rank $r = p + q$, index p, signature $s = p - q$.

2. Main results

Theorem (Sylvester's law of inertia)

- A real symmetric A is congruent to exactly one $\mathrm{diag}(I_p, -I_q, 0)$. Two real forms are equivalent iff they have the same rank and signature.
- Real symmetric $n \times n$ matrices fall into $\frac{(n+1)(n+2)}{2}$ congruence classes; over $\mathbb{C}$, only the rank matters ($n + 1$ classes).
- p is the largest dimension of a subspace on which q is positive definite.
- Bilinear forms on F^n: dimension n^2; symmetric $\frac{n(n+1)}{2}$; alternating $\frac{n(n-1)}{2}$. An alternating form has even rank.

Type	Eigenvalues	Minor test
positive definite	all > 0	leading minors $D_k > 0$
positive semi-definite	all ≥ 0	*all* principal minors ≥ 0
negative definite	all < 0	$(-1)^k D_k > 0$
indefinite	both signs	e.g. $\det < 0$ with n even

Theorem (Values and geometry)

- On $\|x\| = 1$, $x^t A x$ takes exactly the values in $[\lambda_{\min}, \lambda_{\max}]$, the extremes at unit eigenvectors.
- q vanishes at some $v \neq 0$ iff q is indefinite or singular.
- In $\mathbb{R}^2$, $x^t A x = 1$ is an ellipse (A positive definite, semi-axes $1/\sqrt{\lambda_i}$), a hyperbola ($\det A < 0$), or two parallel lines (semi-definite, rank 1).

3. Methods

Method: reducing a form by completing squares (Lagrange).

- (1) Collect the terms with some x_i whose square appears and complete the square.
- (2) Repeat with the remaining variables.
- (3) If no square appears (e.g. $2xy$), put $x = u + v$, $y = u - v$: $2xy = 2u^2 - 2v^2$.
- (4) Count positive and negative squares: these are p and q.

Method: classifying with minors.

- When no leading minor D_k vanishes, q (the number of negative eigenvalues) is the number of sign changes in $1, D_1, \ldots, D_n$.
- If some $D_k = 0$, use eigenvalues or all principal minors.

4. Exam shortcuts and common traps

- $ax^2 + 2bxy + cy^2$: positive definite iff $a > 0$, $ac - b^2 > 0$; indefinite iff $ac - b^2 < 0$.
- $2(xy + yz + zx)$ has matrix $J_3 - I$, eigenvalues $2, -1, -1$: rank 3, signature -1. aa^t is always positive semi-definite of rank 1.
- Only the symmetric part $\frac{1}{2}(A + A^t)$ affects $x^t A x$: $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ gives $x^2 + y^2$.
- **Traps:** forgetting to halve the cross terms; using leading minors to decide semi-definiteness ($\mathrm{diag}(0, -1)$ has $D_1 = D_2 = 0$); confusing congruence (keeps signs) with similarity (keeps eigenvalues); signature $p - q$ is not the index p.

5. Solved examples

Example 1. Classify $q = x^2 + 2y^2 + 3z^2 + 2xy + 2yz$.

Solution

- Matrix $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}$, leading minors $1, 1, 2$.
- All positive: positive definite. Indeed $q = (x + y)^2 + (y + z)^2 + 2z^2$.

Example 2. Find the rank, index and signature of $q = x^2 - y^2 + 2xz$.

Solution

- $q = (x+z)^2 - z^2 - y^2$: one positive and two negative squares.
- Rank 3, index 1, signature -1. (Eigenvalues $-1, \frac{1\pm\sqrt{5}}{2}$.)

Example 3. Find the maximum and minimum of $3x^2 + 2xy + 3y^2$ on $x^2 + y^2 = 1$.

Solution

- Matrix $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ has eigenvalues $4, 2$.
- Maximum 4 at $\pm\frac{1}{\sqrt{2}}(1,1)$; minimum 2 at $\pm\frac{1}{\sqrt{2}}(1,-1)$.

Example 4. Identify the conic $5x^2 + 4xy + 2y^2 = 1$.

Solution

- Matrix $\begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$: $\det = 6 > 0$, trace > 0, eigenvalues $6, 1$.
- An ellipse with semi-axes $1/\sqrt{6}$ and 1.

Example 5. Write $f(x,y) = x_1y_1 + 2x_1y_2 + 4x_2y_1 + 3x_2y_2$ as symmetric plus alternating, and classify $q(x) = f(x,x)$.

Solution

- $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
- $q = x_1^2 + 6x_1x_2 + 3x_2^2$; $ac - b^2 = 3 - 9 < 0$: indefinite.

Example 6. How many congruence classes do 3×3 real symmetric matrices form? Complex symmetric?

Solution

- Real: pairs (p,q) with $p + q \le 3$: $\frac{4\cdot 5}{2} = 10$.
- Complex: one class per rank $0, 1, 2, 3$: 4.